TM Forum Specification

Cost Management User Guide

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Introduction

The following document is the specification of the REST API for Cost management. It includes the model definition as well as all available operations. Possible actions are creating, updating, deleting and retrieving a Cost instance.

The Cost Management API provides a standardized mechanism for creating a cost instance with all the necessary cost attributes. The API consists of a simple set of operations that can interact with various components in the ODA function blocks

There are two types of costs defined in the Cost Management API:

- Projected Costs
  - Projected costs refer to estimated or forecasted expenses that are anticipated to be incurred in the future
- Actual Costs
  - Actual costs represent the real, verified expenses that have been incurred and recorded.

Projected costs are estimates of future expenses made during the planning phase, while actual costs are the realized and verified expenses based on factual data.

A Cost is made up of one or more Cost Items. There are three cycles for Cost Item:

- One Time Cost
- Recurring fixed Cost
- Consumption based cost



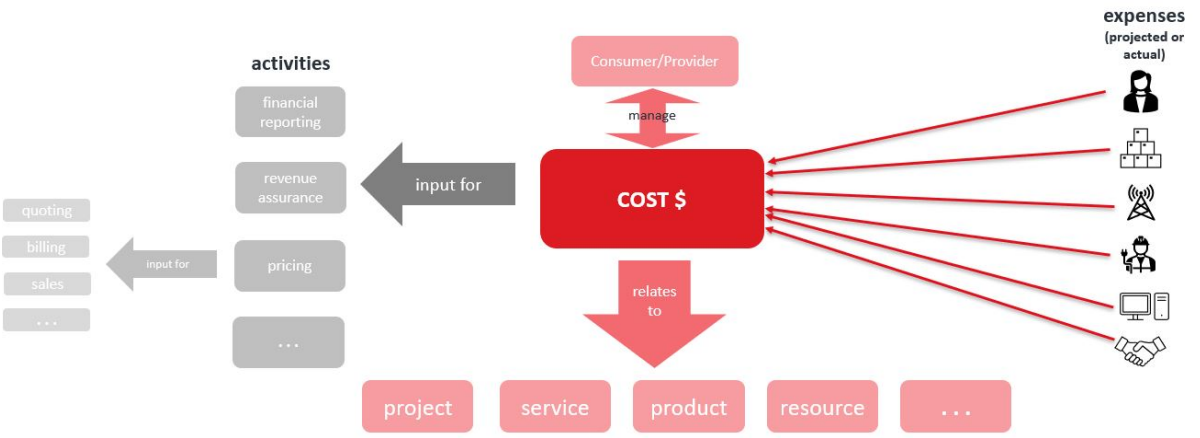
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Use Cases

A cost instance is created to capture the expense (projected or actual) needed to be spent by the service provider to acquire, execute, produce, activate, operate or qualify the entity (project, service, product, resource etc). It encompasses the amount of money (forecasted or incurred) needed in, for example (not exhaustive):

- service fulfilment process, including materials, labour, overhead costs, 3rd party, and any other expenses directly associated with the acquisition of a service
- service qualification process, including materials, labour, overhead costs, and any other expenses directly associated with qualifying a service
- service discovery process, that is, the cost returned to requesting entity for the various candidate target service and resources
- project delivery, including resources time, materials, overhead costs, etc
- resource acquisition process, licenses, materials, labour, overhead costs, etc
- marketing, advertising, sales & disctribution, support staff costs, etc.

In turn, the costs can be used to derive pricing, perform revenue assurance activities and produce financial reports etc.



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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a ProjectedCost and ActualCost inheriting properties from the abstract Cost entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Cost instances some may be instances of ProjectedCost where other could be instances of ActualCost. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an CostRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as ProjectedCostRef or ActualCostRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

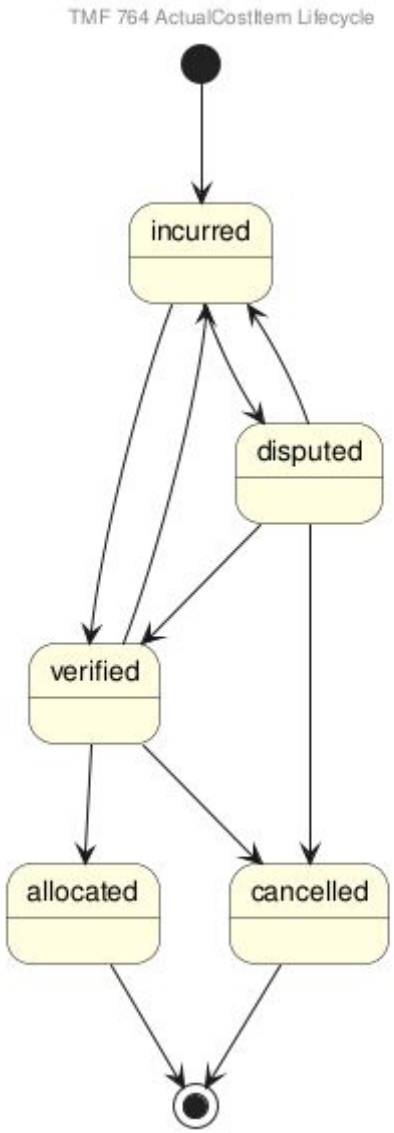
The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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STATE LIFECYCLE

ActualCostItem state lifecycle

Following are the available state values for an actual cost item. The state graphic gives an overview of the allowed state changes



The following table provides the actual cost item state explanation:

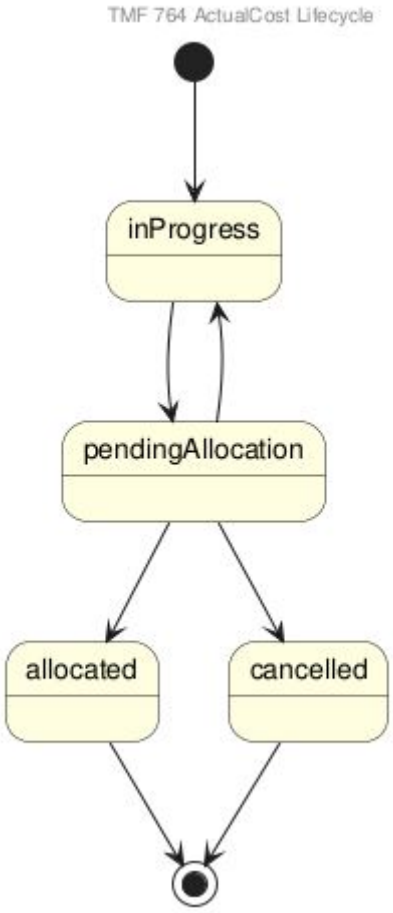
State	Definition
incurred	This is an initial state of an actual cost item that typically occurs when the cost is first acquired, such as when an invoice is received, or a transaction is processed.
verified	This state indicates that an actual cost item has been reviewed, validated, and confirmed as accurate and legitimate. This verification process involves ensuring that the cost aligns with the organization's policies, procedures, and financial records.

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State	Definition
disputed	This state indicates that an actual cost item may not be legitimate according to established rules, regulations or contractual agreements.
allocated	This state indicates that an actual cost item has been assigned or allocated to the appropriate accounting categories, cost centers, projects, or accounts within the organization's financial structure.
cancelled	This state indicates the actual cost item will not be assigned or allocated to the appropriate accounting categories, cost centers etc as the actual cost item has been cancelled.

ActualCost state lifecycle

Following are the available state values for an actual cost. The state graphic gives an overview of the allowed state changes



The following table provides the actual cost state explanation:

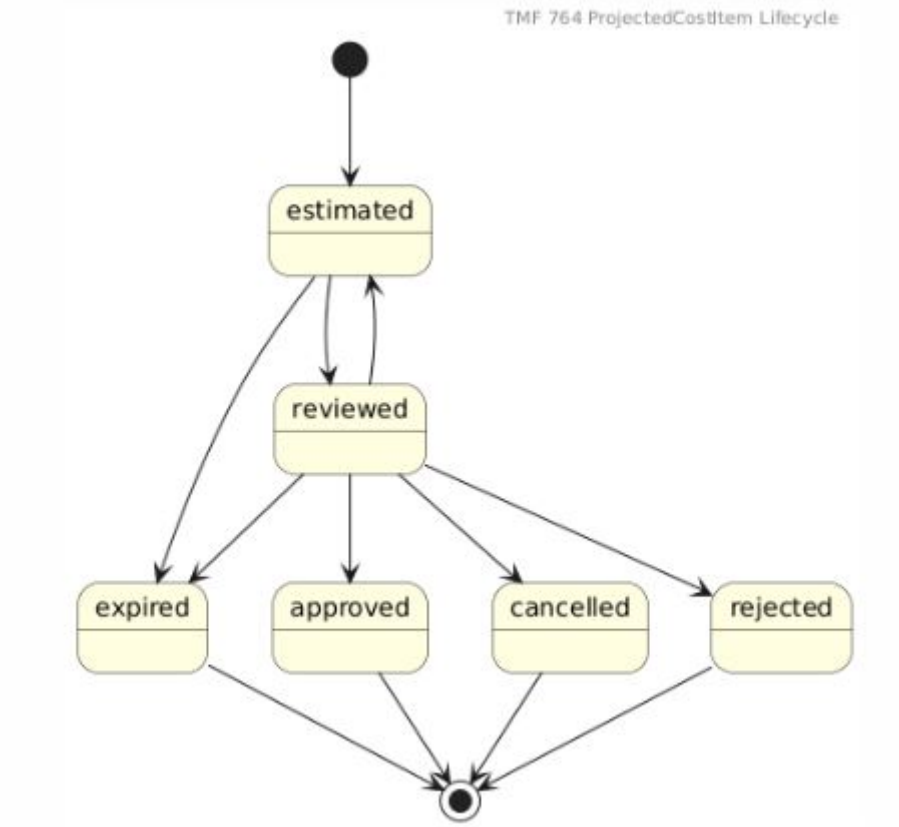
State	Definition
inProgress	The actual cost instance has been generated and some actual cost items are yet to be assoicated to the Actual Cost instance.

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State	Definition
pendingAllocation	All cost items are assoicated to Actual Cost instance. However, not all the actual cost items have reached allocated or cancelled state.
allocated	At least one actual cost item has reached allocated state. The remaining actual cost items are cancelled state.
cancelled	All the actual cost state items have reached cancelled state.

ProjectedCostItem state lifecycle

Following are the available state values for an projected cost item. The state graphic gives an overview of the allowed state changes



The following table provides the acprojectedtual cost item state explanation:

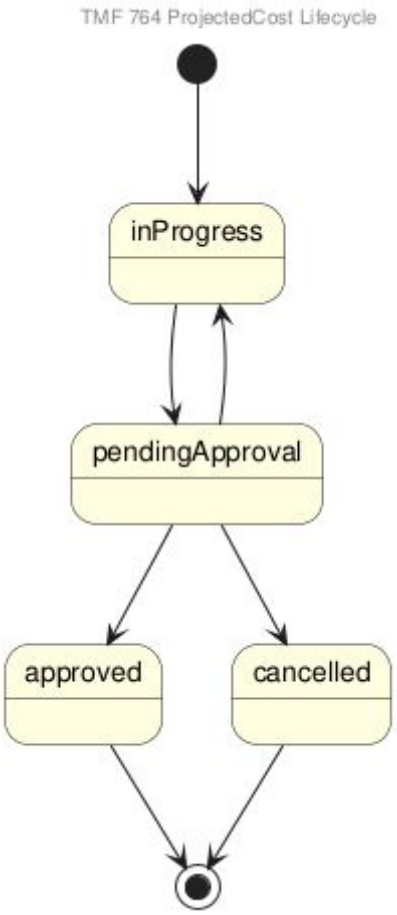
State	Definition
estimated	This state indicates that the expenses for the projected cost item have been estimated by the cost provider and is ready for approval.
reviewed	This state indicates that the expenses for the projected cost item have been reviewed.
approved	This state indicates that the expenses for the projected cost item have been approved.

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State	Definition
rejected	This state indicates that the expenses for the projected cost item have been rejected.
cancelled	This state indicates that the expenses for the projected cost item have been cancelled.
expired	This state indicates the duration for approval/rejection by consuming stakeholder has passed and the projected cost item is expired.

ProjectedCost state lifecycle

Following are the available state values for an projected cost. The state graphic gives an overview of the allowed state changes



The following table provides the actual cost state explanation:

State	Definition
inProgress	The project cost instance has been generated and some project cost items are yet to be assoicated to the Projected Cost instance.
pendingApproval	All cost items are assoicated to Projected Cost instance. Not all the projected cost items have reached a terminal state.

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State	Definition
approved	At least one projected cost item has reached approved state. The remaining projected cost items are expired, rejcted or cancelled state.
cancelled	There are no projected cost items that are approved. The projected cost items are either in expired, rejected or cancelled state.

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RESOURCE MODEL

Managed Entity and Task Resource Models

ActualCost resource

Concrete implementation of Cost, as ActualCost.

Resource model

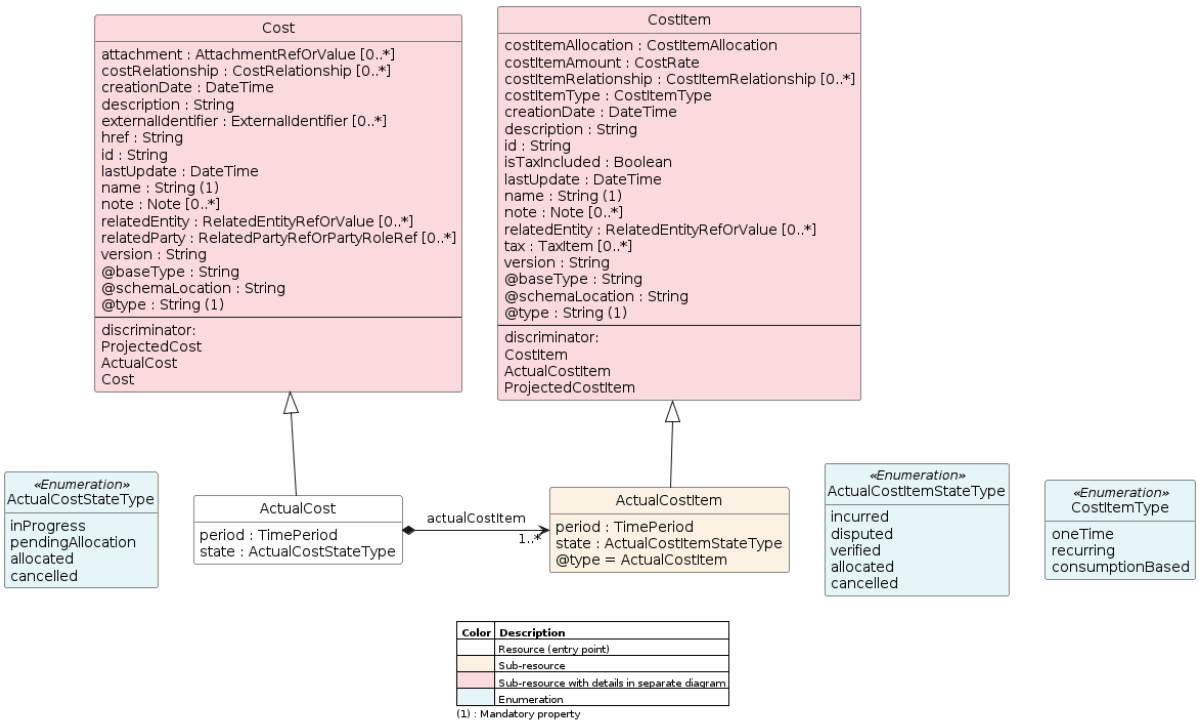


Figure 1 - ActualCost

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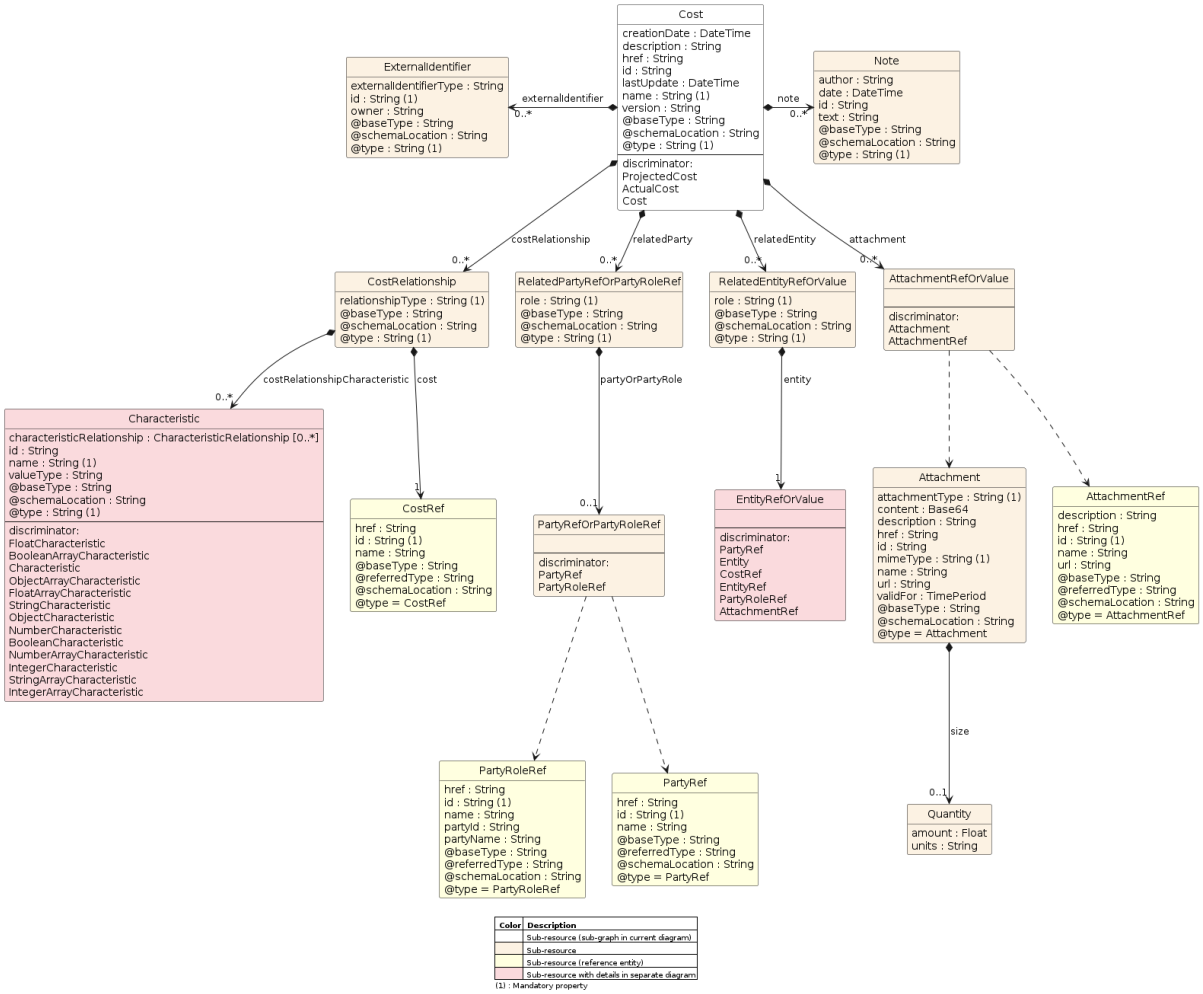


Figure 2 - Cost

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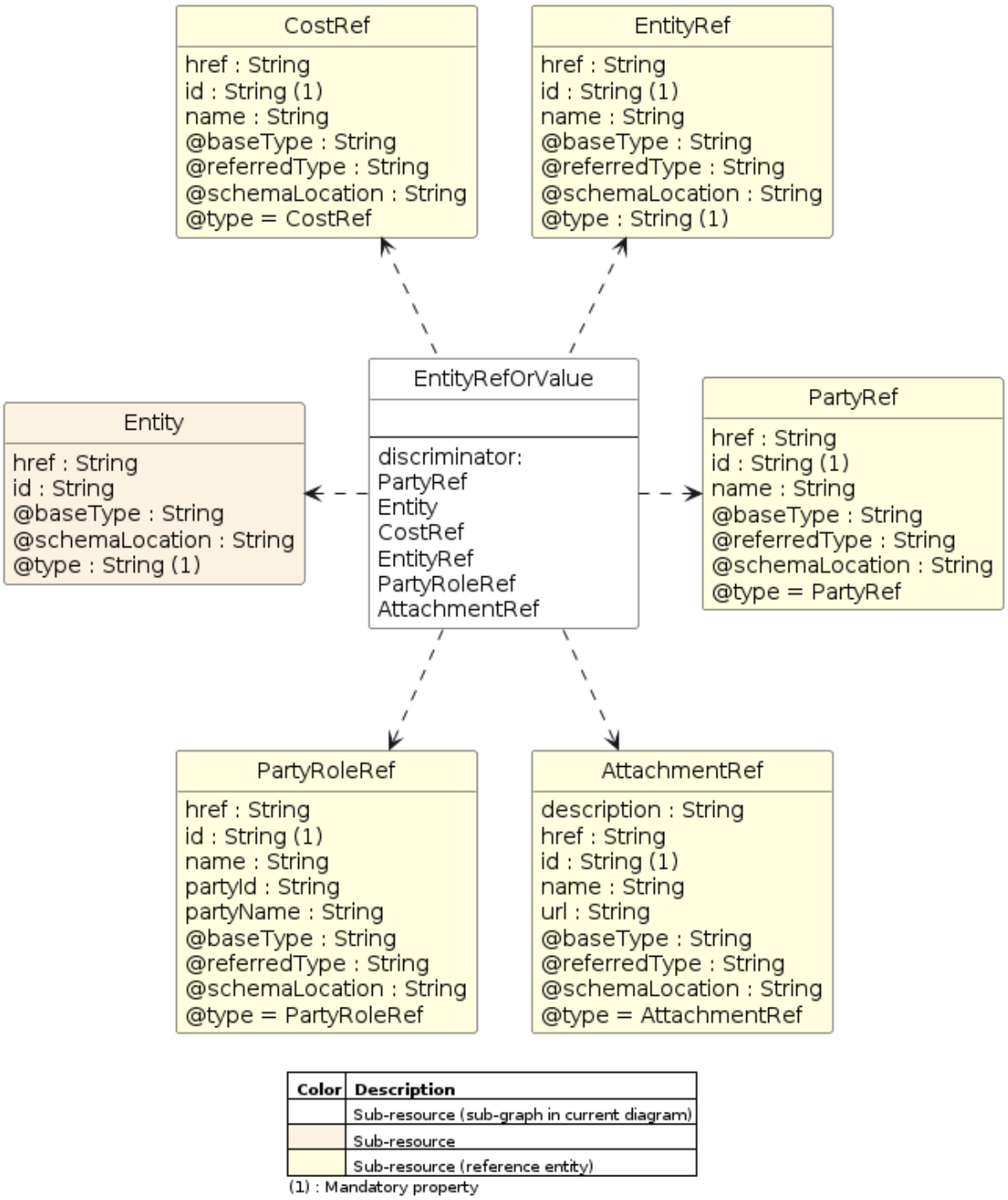


Figure 3 - EntityRefOrValue

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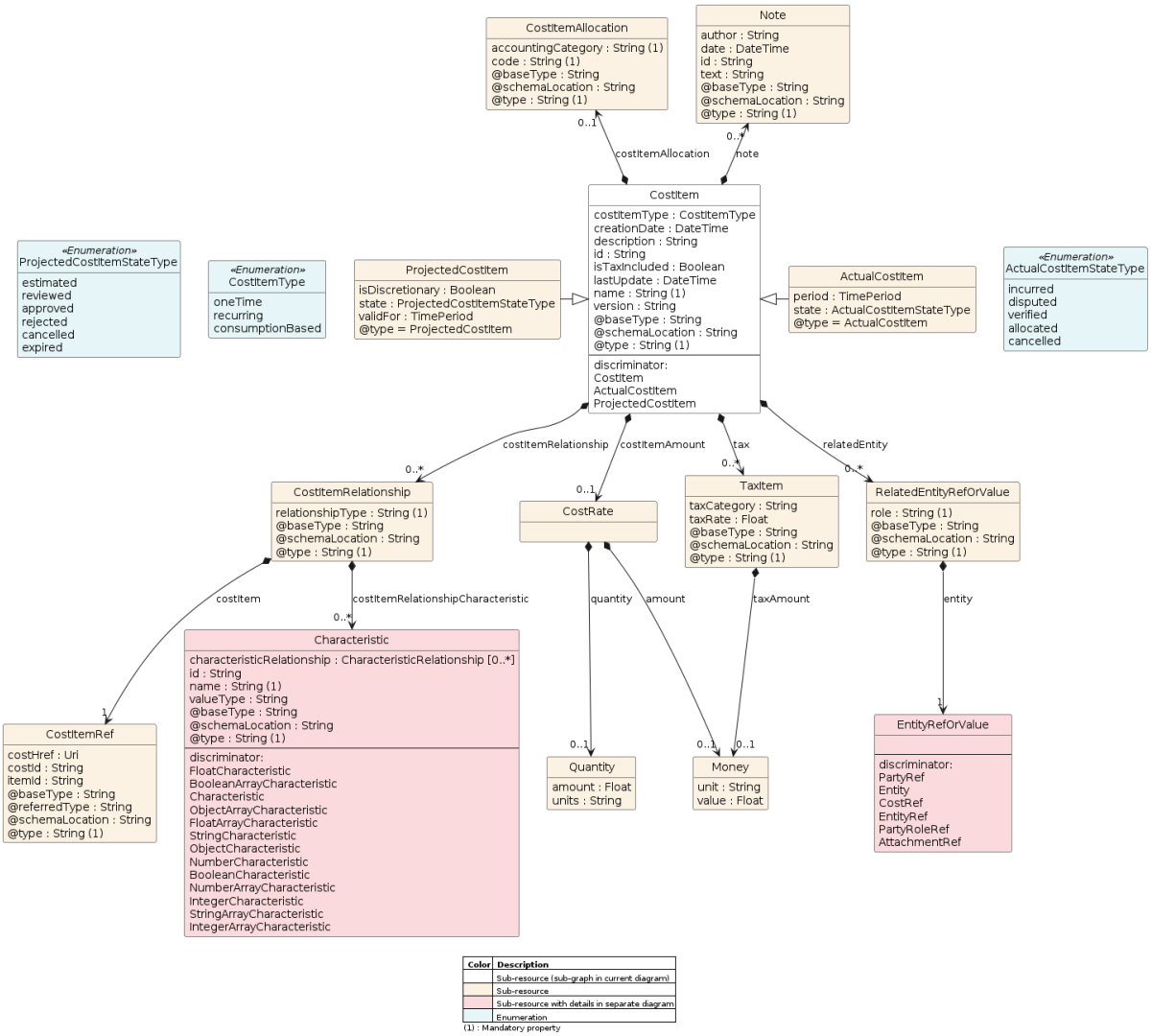


Figure 4 - CostItem

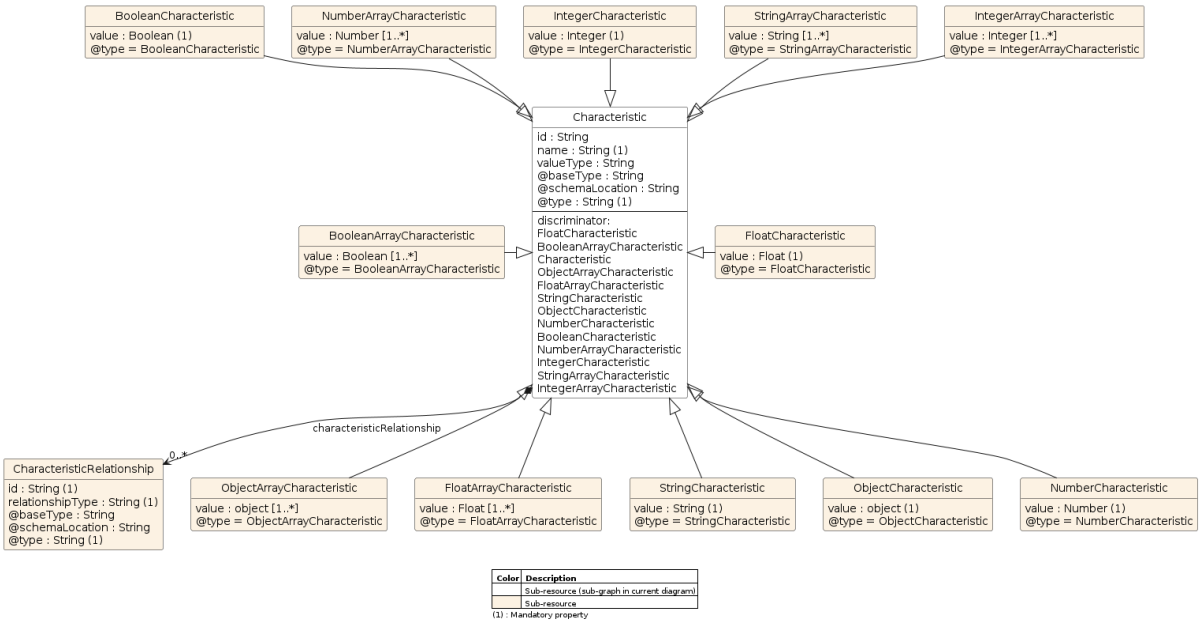


Figure 5 - Characteristic

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Field descriptions

ActualCost fields

actualCostItem	An ActualCostItem. A list of Actual Cost Items items which are part of the overall Actual cost.
attachment	An AttachmentRefOrValue. The attachments pertaining to the cost.
costRelationship	A CostRelationship. A list of cost relationships (CostRelationship [*]). Describes links with other cost(s).
creationDate	A DateTime. The date when the cost object is created.
description	A String. Free-text description of the Cost.
externalIdentifier	An ExternalIdentifier. A list of external identifiers assoicated with this cost.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. The date when the cost object is last updated.
name	A String. Name of the Cost.
note	A Note. A list of notes made on this Cost.
period	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
relatedEntity	A RelatedEntityRefOrValue. A list of related entity in relation to this cost.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of related party references playing a role in this cost object (e.g. - Project Manager, accountant, Procurement Officer, external vendor etc.).
state	An ActualCostStateType. Possible values for the state of the actual cost. ENUMERATED with values: * inProgress * pendingAllocation * allocated * cancelled
version	A String. The version of the Cost.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ActualCostItem sub-resource fields

costItemAllocation	A CostItemAllocation. To indicate under which accounting category the cost item is recorded in the organization.

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costItemAmount	A CostRate. To represent a cost per unit. e.g. $10/hr, 33 pence/min.
costItemRelationship	A CostItemRelationship. A list of cost item relationships (CostRelationship [*]). Describes links with other cost item(s).
costItemType	A CostItemType. Possible values for the cost item. ENUMERATED with values: * oneTime * recurring * consumptionBased
creationDate	A DateTime. The date when the cost item object is created.
description	A String. Free-text description of the Cost Item.
id	A String. A locally unique identifier for the cost item within the list of items.
isTaxIncluded	A Boolean. Indicates whether cost amount includes taxes.
lastUpdate	A DateTime. The date when the cost item object is last updated.
name	A String. Name of the Cost Item.
note	A Note. Extra-information about the cost item.
period	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
relatedEntity	A RelatedEntityRefOrValue. A list of related entity in relation to this cost item.
state	An ActualCostItemStateType. Possible values for the state of the actual cost item. ENUMERATED with values: * incurred * disputed * verified * allocated * cancelled
tax	A TaxItem. An amount of money levied on the cost by a legislative body.
version	A String. The version of the Cost Item.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CostItemAllocation sub-resource fields

accountingCategory	A String. The accounting category within the organization under which the cost item is recorded. This refers to the classification used within the organization to categorize and track cost (e.g.: Cost Centre, Business Unit, etc.).

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code	A String. The code for the accounting category within the organization where the cost is tracked (e.g., code for Cost Centre, Business Unit, etc.).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CostRate sub-resource fields

amount	A Money. A base / value business entity used to represent money.
quantity	A Quantity. An amount in a given unit.

CostItemRelationship sub-resource fields

costItem	A CostItemRef. Reference to a CostItem.
costItemRelationshipCharacteristic	A Characteristic. A list of characteristics of the Cost Item relationship. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
relationshipType	A String. Relationship type as relies on, bundles, etc…
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CostItemRef sub-resource fields

costHref	A Uri. Navigable reference to the cost to which this cost item belongs to.
costId	A String. Identifier of the cost that contains the referred cost item.
itemId	A String. Identifier of the cost item within its parent Cost entity.

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@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedEntityRefOrValue sub-resource fields

entity	An EntityRefOrValue.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CostRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TaxItem sub-resource fields

taxAmount	A Money. A base / value business entity used to represent money.
taxCategory	A String. Tax category.
taxRate	A Float. Applied rate of the tax.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CostItem sub-resource fields

costItemAllocation	A CostItemAllocation. To indicate under which accounting category the cost item is recorded in the organization.
costItemAmount	A CostRate. To represent a cost per unit. e.g. $10/hr, 33 pence/min.
costItemRelationship	A CostItemRelationship. A list of cost item relationships (CostRelationship [*]). Describes links with other cost item(s).
costItemType	A CostItemType. Possible values for the cost item. ENUMERATED with values: * oneTime * recurring * consumptionBased
creationDate	A DateTime. The date when the cost item object is created.
description	A String. Free-text description of the Cost Item.
id	A String. A locally unique identifier for the cost item within the list of items.
isTaxIncluded	A Boolean. Indicates whether cost amount includes taxes.
lastUpdate	A DateTime. The date when the cost item object is last updated.
name	A String. Name of the Cost Item.
note	A Note. Extra-information about the cost item.
relatedEntity	A RelatedEntityRefOrValue. A list of related entity in relation to this cost item.

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tax	A TaxItem. An amount of money levied on the cost by a legislative body.
version	A String. The version of the Cost Item.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	CostItem can be instantiated as * ActualCostItem * ProjectedCostItem
isDiscretionary	This property is present in subclasses
period	This property is present in subclasses
state	This property is present in subclasses
validFor	This property is present in subclasses

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

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See [AttachmentRef] for details.

CostRelationship sub-resource fields

cost	A CostRef. Reference to a Cost entity.
costRelationshipCharacteristic	A Characteristic. A list of characteristics of the Cost relationship. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
relationshipType	A String. Relationship type as relies on, bundles, etc…
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

See [Note] for details.

RelatedEntityRefOrValue sub-resource fields

See [RelatedEntityRefOrValue] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

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partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ActualCost' resource object.


```
{
  "@type": "ActualCost",
  "@baseType": "Cost",
  "@schemaLocation": "http://my.schemas/ActualCost.schema.json",
  "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
  "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
  "name": "Actual Cost",
  "version": "1.0",
  "description": "Costs incurred for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
  "creationDate": "2023-11-14T11:03:08.176Z",
  "lastUpdate": "2024-08-14T11:03:08.176Z",
  "state": "allocated",
  "note": [
    {
      "id": "77456",
      "author": "Jack Smith",
      "date": "2019-05-31T07:34:45.968Z",
      "text": "Any potential upgrades to the service could impact future costs",
      "@type": "Note",
      "@baseType": "Note",
      "@schemaLocation": "http://my.schemas/Note.schema.json"
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives Technical specifications of fibre optic
cable.",
      "mimeType": "jpeg",
      "attachmentType": "TechnicalSpecification",
      "name": "Fibre Cable Specification",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536",
      "size": {
        "amount": 380,
        "units": "KB"
      },
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "@type": "Attachment"
```


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```
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@baseType": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
      "role": "Supplier",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@baseType": "partyOrPartyRole",
        "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "@baseType": "RelatedEntityRefOrValue",
      "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
      "role": "initiator",
      "entity": {
        "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "name": "ServiceOrder",
        "@type": "EntityRef",
        "@baseType": "EntityRef",
        "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
        "@referredType": "ServiceOrder"
      }
    }
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "@baseType": "ExternalIdentifier",
      "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
      "id": "1234",
      "owner": "ChannelPartner",
      "externalIdentifierType": "SupplierInvoiceNumber"
    }
  ],
  "costRelationship": [
    {
      "@type": "CostRelationship",
      "@baseType": "CostRelationship",
      "@schemaLocation": "http://my.schemas/CostRelationship.schema.json",
      "relationshipType": "DependentOn",
      "cost": {
        "@type": "CostRef",
        "id": "58724a75-b436-4371-a8bf-422269599694",
        "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/actualCost/58724a75-b436-4371-a8bf-422269599694"
      },
      "costRelationshipCharacteristic": [
        {
```


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```
          "id": "126",
          "name": "CrossRef",
          "value": "Primary",
          "valueType": "string",
          "characteristicRelationship": [],
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "@schemaLocation": "http://my.schemas/StringCharacteristic.schema.json"
        }
      ]
    }
  ],
  "period": {
    "startDateTime": "2023-11-14T11:03:08.176Z",
    "endDateTime": "2024-08-14T11:03:08.176Z"
  },
  "actualCostItem": [
    {
      "@type": "ActualCostItem",
      "id": "636363",
      "description": "E-line connectivity service",
      "name": "E-line",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAllocation": {
        "accountingCategory": "Business Unit",
        "code": "#789789",
        "@type": "CostItemAllocation"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
          "value": 500
        }
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 20
          }
        }
      ],
      "note": [
        {
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "This cost has been reviewed and verified before allocation",
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "relatedEntity": [
```


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```
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "originator",
          "entity": {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "name": "Service",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
            "@referredType": "Service"
          }
        }
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "898989",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ],
      "state": "allocated",
      "costItemType": "recurring",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      }
    },
    {
      "@type": "ActualCostItem",
      "id": "123456",
      "description": "The actual cost of the fibre optic cable",
      "name": "Fiber Optic Cable",
      "creationDate": "2024-06-14T11:03:08.178Z",
      "lastUpdate": "2024-07-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAllocation": {
        "accountingCategory": "Business Unit",
        "code": "#BU456",
        "@type": "CostItemAllocation"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
        }
```


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```
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 100
          }
        }
      ],
      "note": [
        {
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "Technical specification for the fibre optic cable is provided in
attachment",
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "originator",
          "entity": {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "http://host/resourceOrdering/v5/resourceOrder/3f845e07-73d8-
469b-949f-fe95e52ab5ab",
            "name": "resourceOrder",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
            "@referredType": "resourceOrder"
          }
        }
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "111000",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ],
      "costItemType": "oneTime",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2025-08-14T11:03:08.176Z"
```


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```
      },
      "state": "allocated"
    },
    {
      "@type": "ActualCostItem",
      "id": "234567",
      "description": "The actual cost of the Electricity and Power Consumption",
      "name": "Electricity and Power Consumption",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAllocation": {
        "accountingCategory": "CostCentre",
        "code": "XYZ890",
        "@type": "CostItemAllocation"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 20000,
          "units": "kWh"
        },
        "amount": {
          "unit": "AUD",
          "value": 4000
        }
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 20
          }
        }
      ],
      "note": [
        {
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "The actual cost matches the projected cost for this period",
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "originator",
          "entity": {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/resourceUsagemanagement/v5/resourceUsage/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "name": "ResourceUsage",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
            "@referredType": "ResourceUsage"
          }
        }
```


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```
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "456456",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ],
      "costItemType": "consumptionBased",
      "state": "allocated",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      }
    },
    {
      "@type": "ActualCostItem",
      "id": "234569",
      "description": "Maintenance and support",
      "name": "Maintenance and support",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAllocation": {
        "accountingCategory": "CostCentre",
        "code": "ABC123",
        "@type": "CostItemAllocation"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
          "value": 300
        }
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 20
          }
        }
      ],
      "note": [
        {
```


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```
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "The incurred cost exceeded the projected cost for this period",
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "maintenance",
          "entity": {
            "id": "f8ec1c38-4e4d-4378-a693-f95680e51e25",
            "href": "https://mycsp.com/tmf-
api//workOrderManagement/v5/workOrder/f8ec1c38-4e4d-4378-a693-f95680e51e25",
            "name": "WorkOrder",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
            "@referredType": "WorkOrder"
          }
        }
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "456458",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ],
      "state": "allocated",
      "costItemType": "recurring",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      }
    }
  ]
}
```


ProjectedCost resource

Concrete implementation of Cost, as ProjectedCost.

Resource model

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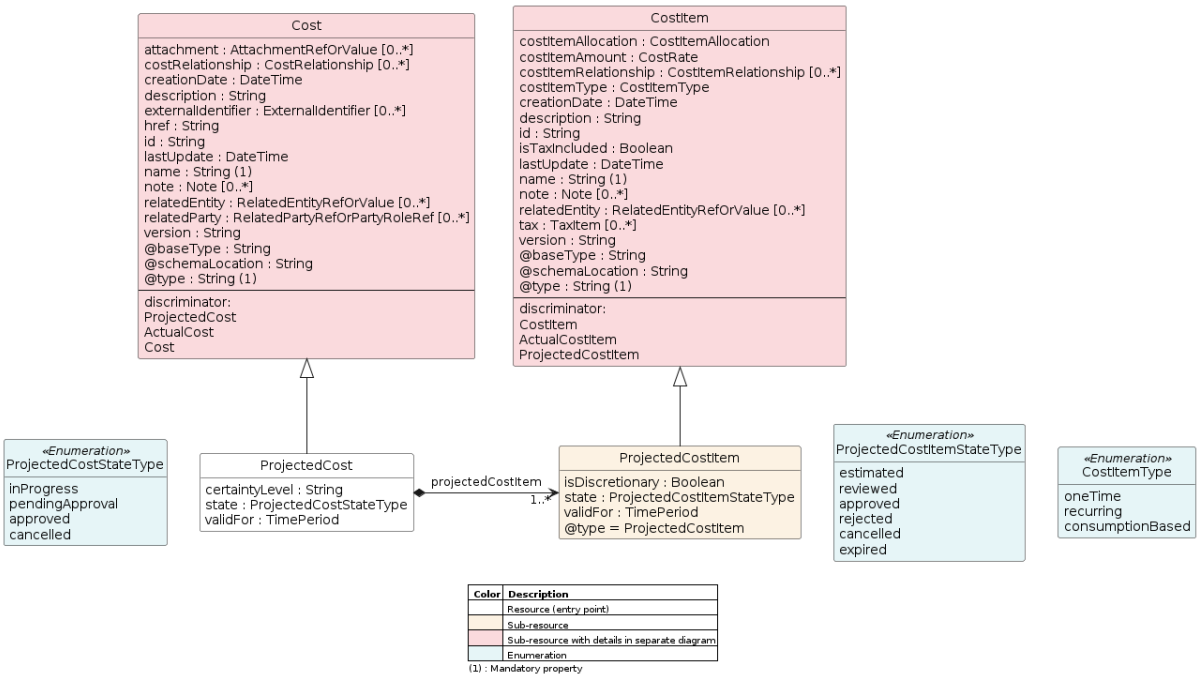


Figure 6 - ProjectedCost

See Figure 2 for Cost diagrams.

See Figure 3 for EntityRefOrValue diagrams.

See Figure 4 for CostItem diagrams.

See Figure 5 for Characteristic diagrams.

Field descriptions

ProjectedCost fields

attachment	An AttachmentRefOrValue. The attachments pertaining to the cost.
certaintyLevel	A String. Indicates the extent of confidence in the accuracy and reliability of the projected cost estimates. Example values can be - 'Indicative cost' an early-stage estimate used to provide a preliminary idea of potential costs. 'Firm cost' also known as fixed cost or committed cost: represents a cost estimate that has undergone detailed analysis, validation, and approval. It is considered highly reliable and stable.
costRelationship	A CostRelationship. A list of cost relationships (CostRelationship [*]). Describes links with other cost(s).
creationDate	A DateTime. The date when the cost object is created.
description	A String. Free-text description of the Cost.
externalIdentifier	An ExternalIdentifier. A list of external identifiers assoicated with this cost.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
lastUpdate	A DateTime. The date when the cost object is last updated.
name	A String. Name of the Cost.
note	A Note. A list of notes made on this Cost.
projectedCostItem	A ProjectedCostItem. A list of Projected Cost Items items which are part of the overall projected cost.
relatedEntity	A RelatedEntityRefOrValue. A list of related entity in relation to this cost.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of related party references playing a role in this cost object (e.g. - Project Manager, accountant, Procurement Officer, external vendor etc.).
state	A ProjectedCostStateType. Possible values for the state of the projected cost. ENUMERATED with values: * inProgress * pendingApproval * approved * cancelled
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. The version of the Cost.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

See [Attachment] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

CostRelationship sub-resource fields

See [CostRelationship] for details.

CostRef sub-resource fields

See [CostRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

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See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

See [FloatArrayCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

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Note sub-resource fields

See [Note] for details.

ProjectedCostItem sub-resource fields

costItemAllocation	A CostItemAllocation. To indicate under which accounting category the cost item is recorded in the organization.
costItemAmount	A CostRate. To represent a cost per unit. e.g. $10/hr, 33 pence/min.
costItemRelationship	A CostItemRelationship. A list of cost item relationships (CostRelationship [*]). Describes links with other cost item(s).
costItemType	A CostItemType. Possible values for the cost item. ENUMERATED with values: * oneTime * recurring * consumptionBased
creationDate	A DateTime. The date when the cost item object is created.
description	A String. Free-text description of the Cost Item.
id	A String. A locally unique identifier for the cost item within the list of items.
isDiscretionary	A Boolean. This indicates discretion or choice to incur the cost item based on preferences, needs, or other constraints. Only discretionary cost could be rejected.
isTaxIncluded	A Boolean. Indicates whether cost amount includes taxes.
lastUpdate	A DateTime. The date when the cost item object is last updated.
name	A String. Name of the Cost Item.
note	A Note. Extra-information about the cost item.
relatedEntity	A RelatedEntityRefOrValue. A list of related entity in relation to this cost item.
state	A ProjectedCostItemStateType. Possible values for the state of the projected cost item. ENUMERATED with values: * estimated * reviewed * approved * rejected * cancelled * expired
tax	A TaxItem. An amount of money levied on the cost by a legislative body.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. The version of the Cost Item.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CostItemAllocation sub-resource fields

See [CostItemAllocation] for details.

CostRate sub-resource fields

See [CostRate] for details.

CostItemRelationship sub-resource fields

See [CostItemRelationship] for details.

CostItemRef sub-resource fields

See [CostItemRef] for details.

RelatedEntityRefOrValue sub-resource fields

See [RelatedEntityRefOrValue] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

TaxItem sub-resource fields

See [TaxItem] for details.

CostItem sub-resource fields

See [CostItem] for details.

RelatedEntityRefOrValue sub-resource fields

See [RelatedEntityRefOrValue] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'ProjectedCost' resource object.

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```
{
  "@type": "ProjectedCost",
  "@baseType": "Cost",
  "@schemaLocation": "http://my.schemas/ProjectedCost.schema.json",
  "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/projectedCost/19d72f32-
13b8-4711-a805-12850ea1ce20",
  "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
  "description": "Cost projection for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
  "name": "Projected Cost",
  "version": "1.0",
  "creationDate": "2024-08-15T14:20:34.396Z",
  "lastUpdate": "2024-08-15T14:20:34.396Z",
  "state": "pendingApproval",
  "note": [
    {
      "id": "77456",
      "author": "Jack Smith",
      "date": "2019-05-31T07:34:45.968Z",
      "text": "Any potential upgrades to the service could impact future costs",
      "@type": "Note",
      "@baseType": "Note",
      "@schemaLocation": "http://my.schemas/Note.schema.json"
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives Technical specifications of fibre optic
cable.",
      "mimeType": "jpeg",
      "attachmentType": "TechnicalSpecification",
      "name": "Fibre Cable Specification",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536",
      "size": {
        "amount": 380,
        "units": "KB"
      },
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "@type": "Attachment"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@baseType": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
      "role": "Supplier",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@baseType": "partyOrPartyRole",
        "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
```


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```
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "@baseType": "RelatedEntityRefOrValue",
      "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
      "role": "initiator",
      "entity": {
        "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "name": "ServiceOrder",
        "@type": "EntityRef",
        "@baseType": "EntityRef",
        "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
        "@referredType": "ServiceOrder"
      }
    }
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "@baseType": "ExternalIdentifier",
      "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
      "id": "1234",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Quotation"
    }
  ],
  "certaintyLevel": "Firm cost",
  "validFor": {
    "startDateTime": "2024-08-15T14:20:34.396Z",
    "endDateTime": "2024-09-15T14:20:34.396Z"
  },
  "costRelationship": [
    {
      "@type": "CostRelationship",
      "@baseType": "CostRelationship",
      "@schemaLocation": "http://my.schemas/CostRelationship.schema.json",
      "relationshipType": "DependentOn",
      "cost": {
        "@type": "CostRef",
        "id": "58724a75-b436-4371-a8bf-422269599694",
        "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/58724a75-b436-4371-a8bf-422269599694"
      },
      "costRelationshipCharacteristic": [
        {
          "id": "126",
          "name": "CrossRef",
          "value": "Primary",
          "valueType": "string",
          "characteristicRelationship": [],
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "@schemaLocation": "http://my.schemas/StringCharacteristic.schema.json"
        }
      ]
    }
  ],
  "projectedCostItem": [
    {
      "@type": "ProjectedCostItem",
```


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```
      "@baseType": "CostItem",
      "@schemaLocation": "http://my.schemas/ProjectedCostItem.schema.json",
      "id": "898989",
      "description": "E-line connectivity service",
      "name": "E-line",
      "version": "1.0",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
      "isTaxIncluded": true,
      "costItemAllocation": {
        "@type": "CostItemAllocation",
        "accountingCategory": "Cost Centre",
        "code": "7897789"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
          "value": 500
        }
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 20
          }
        }
      ],
      "note": [
        {
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "The projection is expected to remain unchanged after review",
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "originator",
          "entity": {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "name": "Service",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
```


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```
            "@referredType": "Service"
          }
        }
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "7cafe309-c089-456f-aa64-8de0c84c9519",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/8cde96d6-6c6f-4254-84bb-bcd24d647dbb",
            "costId": "8cde96d6-6c6f-4254-84bb-bcd24d647dbb"
          },
          "relationshipType": "dependentOn"
        }
      ],
      "costItemType": "recurring",
      "isDiscretionary": false
    },
    {
      "@type": "ProjectedCostItem",
      "id": "111000",
      "description": "The projected cost of the fibre optic cable",
      "name": "Fiber Optic Cable",
      "version": "1.0",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
      "isTaxIncluded": true,
      "costItemAllocation": {
        "@type": "CostItemAllocation",
        "accountingCategory": "Cost Centre",
        "code": "7897789"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
        }
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 100
          }
        }
      ],
      "note": [
        {
```


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```
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "Technical specification for the fibre optic cable is provided in
attachment",
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T23:20:50.52Z",
        "endDateTime": "2024-09-15T23:20:50.52Z"
      },
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "originator",
          "entity": {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "http://host/resourceOrdering/v5/resourceOrder/3f845e07-73d8-
469b-949f-fe95e52ab5ab",
            "name": "resourceOrder",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
            "@referredType": "ResourceOrder"
          }
        }
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "03293842-9b07-4dc2-966f-e69f70b9ead2",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/8cde96d6-6c6f-4254-84bb-bcd24d647dbb",
            "costId": "8cde96d6-6c6f-4254-84bb-bcd24d647dbb"
          },
          "relationshipType": "dependentOn"
        }
      ],
      "costItemType": "oneTime",
      "isDiscretionary": false
    },
    {
      "@type": "ProjectedCostItem",
      "@baseType": "CostItem",
      "@schemaLocation": "http://my.schemas/ProjectedCostItem.schema.json",
      "id": "456456",
      "description": "The projected cost of the Electricity and Power Consumption",
      "name": "Electricity and Power Consumption",
      "version": "1.0",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
```


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```
      "isTaxIncluded": true,
      "costItemAllocation": {
        "@type": "CostItemAllocation",
        "accountingCategory": "Cost Centre",
        "code": "7897789"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 10000,
          "units": "kWh"
        },
        "amount": {
          "unit": "AUD",
          "value": 2000
        }
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 20
          }
        }
      ],
      "note": [
        {
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "The projection is formulated based on the average consumtion
recorded",
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "originator",
          "entity": {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/resourceUsagemanagement/v5/resourceUsage/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "name": "ResourceUsage",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
            "@referredType": "ResourceUsage"
          }
        }
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
```


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```
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "5d1b0768-7912-44bd-93ad-136f64e17e16",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/8cde96d6-6c6f-4254-84bb-bcd24d647dbb",
            "costId": "8cde96d6-6c6f-4254-84bb-bcd24d647dbb"
          },
          "relationshipType": "dependentOn"
        }
      ],
      "costItemType": "consumptionBased",
      "isDiscretionary": false
    },
    {
      "@type": "ProjectedCostItem",
      "@baseType": "CostItem",
      "@schemaLocation": "http://my.schemas/ProjectedCostItem.schema.json",
      "id": "456458",
      "description": "Maintenance and support",
      "name": "Maintenance and support",
      "version": "1.0",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
      "isTaxIncluded": true,
      "costItemAllocation": {
        "@type": "CostItemAllocation",
        "accountingCategory": "Cost Centre",
        "code": "7897789"
      },
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
          "value": 200
        }
      },
      "tax": [
        {
          "taxCategory": "GST",
          "taxRate": 0.1,
          "@type": "TaxItem",
          "taxAmount": {
            "unit": "AUD",
            "value": 20
          }
        }
      ],
      "note": [
        {
          "id": "77457",
          "author": "John",
          "date": "2019-05-31T07:34:45.968Z",
          "text": "This projection is based on standard maintenance tasks; any non-
standard maintenance work may lead to variations in the actual cost",
```


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```
          "@type": "Note",
          "@baseType": "Note",
          "@schemaLocation": "http://my.schemas/Note.schema.json"
        }
      ],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "maintenance",
          "entity": {
            "id": "f8ec1c38-4e4d-4378-a693-f95680e51e25",
            "href": "https://mycsp.com/tmf-
api//workOrderManagement/v5/workOrder/f8ec1c38-4e4d-4378-a693-f95680e51e25",
            "name": "WorkOrder",
            "@type": "EntityRef",
            "@baseType": "EntityRef",
            "@schemaLocation": "http://my.schemas/EntityRef.schema.json",
            "@referredType": "WorkOrder"
          }
        }
      ],
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "6454b64a-639d-4d32-942c-6ed0a14517a6",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/8cde96d6-6c6f-4254-84bb-bcd24d647dbb",
            "costId": "8cde96d6-6c6f-4254-84bb-bcd24d647dbb"
          },
          "relationshipType": "dependentOn"
        }
      ],
      "costItemType": "recurring",
      "isDiscretionary": false
    }
  ]
}
```


Notification Resource Models

8 notifications are defined for this API.

Notifications related to ActualCost:

- Create Event
- Delete Event

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- Attribute Value Change Event
- State Change Event

Notifications related to ProjectedCost:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

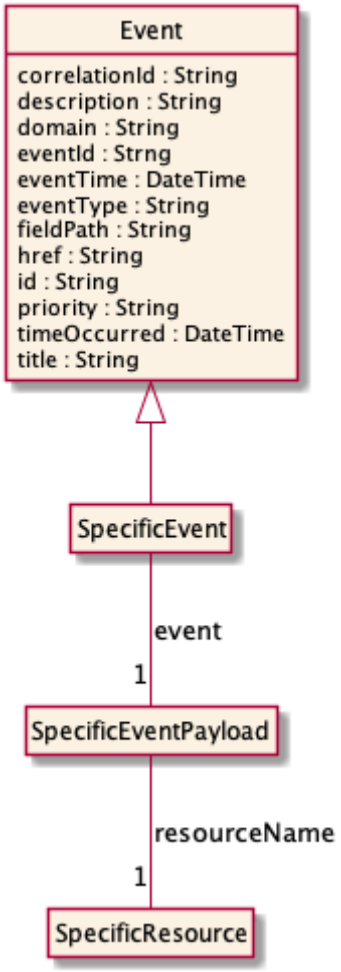


Figure 7 Notification Pattern

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ActualCost

Create Event

Message example for ActualCostCreateEvent event


```
Content-Type: application/json
{
  "correlationId": "2cdbfa6e-6756",
  "description": "ActualCostCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "41af-bb8d-b056f61b8696",
  "eventTime": "2024-08-14T11:03:08.176Z",
  "eventType": "ActualCostCreateEvent",
  "priority": "5",
  "timeOccurred": "2024-08-14T11:03:08.176Z",
  "title": "ActualCostCreateEvent",
  "event": {
    "actualCost": {
      "@type": "ActualCost",
      "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
      "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
      "name": "Actual Cost",
      "creationDate": "2023-11-14T11:03:08.176Z",
      "lastUpdate": "2023-11-14T11:03:08.176Z",
      "state": "inProgress",
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "Supplier",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
            "id": "456",
            "name": "NetCo Pty Ltd",
            "@referredType": "Organization"
          }
        }
      ],
      "relatedEntity": [
        {
          "@type": "RelatedEntityRefOrValue",
          "role": "initiator",
          "entity":
            {
              "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
              "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
              "@type": "EntityRef",
              "@referredType": "ServiceOrder"
            }
        }
      ],
      "period": {
        "startDateTime": "2023-11-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      },
      "actualCostItem": [
        {
```


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```
          "@type": "ActualCostItem",
          "id": "123456",
          "description": "The actual cost of the fibre optic cable",
          "name": "Fiber Optic Cable",
          "creationDate": "2024-06-14T11:03:08.178Z",
          "lastUpdate": "2024-07-14T11:03:08.178Z",
          "isTaxIncluded": true,
          "costItemAmount": {
            "quantity": {
              "amount": 10,
              "units": "km"
            },
            "amount": {
              "unit": "AUD",
              "value": 50000
            }
          },
          "state": "incurred",
          "costItemRelationship": [
            {
              "@type": "CostItemRelationship",
              "@baseType": "CostItemRelationship",
              "@schemaLocation":
"http://my.schemas/CostItemRelationship.schema.json",
              "costItemRelationshipCharacteristic": [],
              "costItem": {
                "@type": "CostItemRef",
                "@baseType": "CostItemRef",
                "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
                "@referredType": "projectedCostItem",
                "itemId": "898989",
                "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
                "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
              },
              "relationshipType": "projectedBy"
            }
          ]
        }
      ]
    }
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ActualCostCreateEvent"
}
```


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Delete Event

Message example for ActualCostDeleteEvent event

```
Content-Type: application/json
{
    "correlationId": "36af9e61-4a51",
    "description": "ActualCostDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4154-a76e-bf08c4bccb2e",
    "eventTime": "2024-08-14T11:03:08.176Z",
    "eventType": "ActualCostDeleteEvent",
    "priority": "1",
    "timeOccurred": "2024-08-14T11:03:08.176Z",
    "title": "ActualCostDeleteEvent",
    "event": {
        "actualCost": {
            "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
            "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/actualCost/19d72f32-13b8-4711-a805-12850ea1ce19",
            "@type": "ActualCostDeleteEvent"
        }
    },
    "reportingSystem": {
        "id": "819",
        "name": "APP-913",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
      },
      "source": {
        "id": "152",
        "name": "APP-335",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
      },
    "@baseType": "Event",
    "@type": "ActualCostDeleteEvent"
}
```

Attribute Value Change Event

Message example for ActualCostAttributeValueChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "b1995ca7-9d12",
  "description": "ActualCostAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4f3f-849c-c2ad0d10fbe1",
  "eventTime": "2024-08-14T11:03:08.176Z",
  "eventType": "AcutalCostAttributeValueChangeEvent",
  "priority": "2",
  "timeOccurred": "2024-08-14T11:03:08.176Z",
  "title": "ActualCostAttributeValueChangeEvent",
  "event": {
    "actualCost": {
      "@type": "ActualCost",
      "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
```

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```
13b8-4711-a805-12850ea1ce19",
      "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
      "externalIdentifier": [
        {
          "@type": "ExternalIdentifier",
          "@baseType": "ExternalIdentifier",
          "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
          "id": "FIN-12345",
          "owner": "AuditManagement",
          "externalIdentifierType": "Auditing Report"
        }
      ]
    }
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ActualCostAttributeValueChangeEvent"
}
```


State Change Event

Message example for ActualCostStateChangeEvent event


```
Content-Type: application/json
{
  "correlationId": "b1995ca7-9d12",
  "description": "ActualCostStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4f3f-849c-c2ad0d10fbe1",
  "eventTime": "2024-08-14T11:03:08.176Z",
  "eventType": "AcutalCostStateChangeEvent",
  "priority": "2",
  "timeOccurred": "2024-08-14T11:03:08.176Z",
  "title": "ActualCostStateChangeEvent",
  "event": {
    "actualCost": {
      "@type": "ActualCost",
      "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
      "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
      "state": "pendingAllocation"
    }
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
```


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```
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ActualCostStateChangeEvent"
}
```

ProjectedCost

Create Event

Message example for ProjectedCostCreateEvent event

```
Content-Type: application/json
{
    "correlationId": "2cdbfa6e-6756",
    "description": "ProjectedCostCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "41af-bb8d-b056f61b8696",
    "eventTime": "2024-08-14T11:03:08.176Z",
    "eventType": "ProjectedCostCreateEvent",
    "priority": "5",
    "timeOccurred": "2024-08-14T11:03:08.176Z",
    "title": "ProjectedCostCreateEvent",
    "event": {
      "projectedCost": {
        "@type": "ProjectedCost",
        "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
        "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
        "description": "Cost projection for Service Order SO1",
        "name": "Projected Cost",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "state": "inProgress",
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Supplier",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
              "id": "456",
              "name": "NetCo Pty Ltd",
              "@referredType": "Organization"
            }
          }
        ],
        "relatedEntity": [
          {
            "@type": "RelatedEntityRefOrValue",
            "role": "initiator",
            "entity":
              {
                "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
```

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```
                "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
                "@type": "EntityRef",
                "@referredType": "servoceOrder"
              }
            }
          ],
          "certaintyLevel": "Firm cost",
          "validFor": {
            "startDateTime": "2024-08-15T14:20:34.396Z",
            "endDateTime": "2024-09-15T14:20:34.396Z"
          },
          "projectedCostItem": [
            {
              "@type": "ProjectedCostItem",
              "id": "111000",
              "description": "The projected cost of the fibre optic cable",
              "name": "Fiber Optic Cable",
              "creationDate": "2024-08-15T14:20:34.396Z",
              "lastUpdate": "2024-08-15T14:20:34.396Z",
              "isTaxIncluded": true,
              "costItemAmount": {
                "quantity": {
                  "amount": 10,
                  "units": "km"
                },
                "amount": {
                  "unit": "AUD",
                  "value": 50000
                }
              },
              "state": "estimated",
              "validFor": {
                "startDateTime": "2024-08-15T23:20:50.52Z",
                "endDateTime": "2024-09-15T23:20:50.52Z"
              },
              "costItemType": "consumptionBased"
            },
            {
              "@type": "ProjectedCostItem",
              "id": "456456",
              "description": "The projected cost of the Electricity and Power
Consumption",
              "name": "Electricity and Power Consumption",
              "creationDate": "2024-08-15T14:20:34.396Z",
              "lastUpdate": "2024-08-15T14:20:34.396Z",
              "isTaxIncluded": true,
              "costItemAmount": {
                "quantity": {
                  "amount": 10000,
                  "units": "kWh"
                },
                "amount": {
                  "unit": "AUD",
                  "value": 2000
                }
              },
              "state": "estimated",
              "validFor": {
                "startDateTime": "2024-08-15T14:20:34.396Z",
                "endDateTime": "2024-09-15T14:20:34.396Z"
              },
              "costItemType": "recurring"
```


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```
          }
        ]
      }
    },
    "reportingSystem": {
      "id": "819",
      "name": "APP-913",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "152",
      "name": "APP-335",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProjectedCostCreateEvent"
  }
```


Delete Event

Message example for ProjectedCostDeleteEvent event


```
Content-Type: application/json
{
    "correlationId": "36af9e61-4a51",
    "description": "ProjectedCostDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4154-a76e-bf08c4bccb2e",
    "eventTime": "2024-08-14T11:03:08.176Z",
    "eventType": "ProjectedCostDeleteEvent",
    "priority": "1",
    "timeOccurred": "2024-08-14T11:03:08.176Z",
    "title": "ProjectedCostDeleteEvent",
    "event": {
        "projectedCost": {
            "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
            "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "@type": "ProjectedCostDeleteEvent"
        }
    },
    "reportingSystem": {
        "id": "819",
        "name": "APP-913",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
      },
      "source": {
        "id": "152",
        "name": "APP-335",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
      },
    "@baseType": "Event",
    "@type": "ProjectedCostDeleteEvent"
}
```


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Attribute Value Change Event

Message example for ProjectedCostAttributeValueChangeEvent event


```
Content-Type: application/json
{
    "correlationId": "2cdbfa6e-6756",
    "description": "ProjectedCostAttributeValueChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "41af-bb8d-b056f61b8696",
    "eventTime": "2024-08-14T11:03:08.176Z",
    "eventType": "ProjectedCostAttributeValueChangeEvent",
    "priority": "5",
    "timeOccurred": "2024-08-14T11:03:08.176Z",
    "title": "ProjectedCostAttributeValueChangeEvent",
    "event": {
      "projectedCost": {
        "@type": "ProjectedCost",
        "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
        "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
        "externalIdentifier": [
          {
            "id": "FIN-12345",
            "externalIdentifierType": "Auditing Report",
            "@type": "ExternalIdentifier",
            "@baseType": "ExternalIdentifier",
            "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json"
          }
        ]
      }
    },
    "reportingSystem": {
      "id": "819",
      "name": "APP-913",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "152",
      "name": "APP-335",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProjectedCostAttributeValueChangeEvent"
  }
```


State Change Event

Message example for ProjectedCostStateChangeEvent event


```
Content-Type: application/json
{
    "correlationId": "2cdbfa6e-6756",
    "description": "ProjectedCostStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "41af-bb8d-b056f61b8696",
```


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```
    "eventTime": "2024-08-14T11:03:08.176Z",
    "eventType": "ProjectedCostStateChangeEvent",
    "priority": "5",
    "timeOccurred": "2024-08-14T11:03:08.176Z",
    "title": "ProjectedCostStateChangeEvent",
    "event": {
      "projectedCost": {
        "@type": "ProjectedCost",
        "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
        "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
        "state": "approved"
      }
    },
    "reportingSystem": {
      "id": "819",
      "name": "APP-913",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "152",
      "name": "APP-335",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProjectedCostStateChangeEvent"
  }
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ActualCost

List or find ActualCost objects

`GET /actualCost?fields=…`

Description

This operation list ActualCost entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple actual cost instances, in this example allocated actual cost instances are filtered and id, href, name and state fields need to be returned

Request

```
GET /actualCost?state=allocated&fields=id,name,state,href
Content-Type: application/json
```

Response

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```
200
[
  {
    "name": "Actual Cost",
    "id":"19d72f32-13b8-4711-a805-12850ea1ce19",
    "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
    "state":"allocated",
    "@type": "ActualCost"
  },
  {
    "name": "Actual Cost",
    "id":"19d72f32-13b8-4711-a805-12850ea89890",
    "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea89890",
    "state":"allocated",
    "@type": "ActualCost"
  }
]
```

Retrieves a ActualCost by ID

`GET /actualCost/{id}?fields=…`

Description

This operation retrieves an ActualCost entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific actual cost instance.

Request

```
GET /actualCost/19d72f32-13b8-4711-a805-12850ea1ce19
Content-Type: application/json
```

Response

```
200
{
  "@type": "ActualCost",
  "@baseType": "Cost",
  "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
  "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
  "name": "Actual Cost",
  "description": "Costs incurred for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
  "creationDate": "2023-11-14T11:03:08.176Z",
  "lastUpdate": "2024-08-14T11:03:08.176Z",
  "state": "allocated",
```

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```
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Supplier",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "role": "initiator",
      "entity": {
        "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "@type": "EntityRef",
        "@referredType": "ServiceOrder"
      }
    }
  ],
  "period": {
    "startDateTime": "2023-11-14T11:03:08.176Z",
    "endDateTime": "2024-08-14T11:03:08.176Z"
  },
  "actualCostItem": [
    {
      "@type": "ActualCostItem",
      "id": "636363",
      "description": "E-line connectivity service",
      "name": "E-line",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
          "value": 500
        }
      },
      "tax": [{
        "taxCategory": "GST",
        "taxRate": 0.10,
        "@type":"TaxItem"
      }],
      "state": "incurred",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      },
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
```


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```
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "898989",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    },
    {
      "@type": "ActualCostItem",
      "id": "123456",
      "description": "The actual cost of the fibre optic cable",
      "name": "Fiber Optic Cable",
      "creationDate": "2024-06-14T11:03:08.178Z",
      "lastUpdate": "2024-07-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
        }
      },
      "tax": [{
        "taxCategory": "GST",
        "taxRate": 0.10,
        "@type":"TaxItem"
      }],
      "costItemType":"oneTime",
      "state": "allocated",
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "111000",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    },
    {
```


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```
      "@type": "ActualCostItem",
      "id": "234567",
      "description": "The actual cost of the Electricity and Power Consumption",
      "name": "Electricity and Power Consumption",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 20000,
          "units": "kWh"
        },
        "amount": {
          "unit": "AUD",
          "value": 4000
        }
      },
      "tax": [{
        "taxCategory": "GST",
        "taxRate": 0.10,
        "@type":"TaxItem"
      }],
      "costItemType":"consumptionBased",
      "state": "incurred",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      },
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "456456",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    },
    {
      "@type": "ActualCostItem",
      "id": "234569",
      "description": "Maintenance and support",
      "name": "Maintenance and support",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
```


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```
            "value": 200
          }
        },
        "tax": [{
          "taxCategory": "GST",
          "taxRate": 0.10,
          "@type":"TaxItem"
        }],
        "state": "incurred",
        "period": {
          "startDateTime": "2024-07-14T11:03:08.176Z",
          "endDateTime": "2024-08-14T11:03:08.176Z"
        },
        "costItemRelationship": [
          {
            "@type": "CostItemRelationship",
            "@baseType": "CostItemRelationship",
            "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
            "costItemRelationshipCharacteristic": [],
            "costItem": {
              "@type": "CostItemRef",
              "@baseType": "CostItemRef",
              "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
              "@referredType": "projectedCostItem",
              "itemId": "456458",
              "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
              "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
            },
            "relationshipType": "projectedBy"
          }
        ]
      }
    ]
}
```


Creates a ActualCost

`POST /actualCost?fields=…`

Description

This operation creates an ActualCost entity.

Mandatory Attributes

Mandatory Attributes	Rule
actualCostItem	
name	
@type	

Usage samples

Here's an example of a request for creating an actual cost

Request

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```
POST /actualCost
Content-Type: application/json
{
  "@type": "ActualCost",
  "@baseType": "Cost",
  "name": "Actual Cost",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Supplier",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "role": "initiator",
      "entity":
        {
          "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "@type": "EntityRef",
          "@referredType": "ServiceOrder"
        }
    }
  ],
  "period": {
    "startDateTime": "2023-11-14T11:03:08.176Z",
    "endDateTime": "2024-08-14T11:03:08.176Z"
  },
  "actualCostItem": [
    {
      "@type": "ActualCostItem",
      "id": "123456",
      "description": "The actual cost of the fibre optic cable",
      "name": "Fiber Optic Cable",
      "creationDate": "2024-06-14T11:03:08.178Z",
      "lastUpdate": "2024-07-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
        }
      },
      "state": "incurred",
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
```

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```
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "111000",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    }
  ]
}
```

Response

```
201
{
  "@type": "ActualCost",
  "@baseType": "Cost",
  "id":"19d72f32-13b8-4711-a805-12850ea1ce19",
  "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
  "name": "Actual Cost",
  "creationDate": "2023-11-14T11:03:08.176Z",
  "lastUpdate": "2023-11-14T11:03:08.176Z",
  "state": "inProgress",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Supplier",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "role": "initiator",
      "entity":
        {
          "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "@type": "EntityRef",
          "@referredType": "ServiceOrder"
        }
    }
  ],
```

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```
  "period": {
    "startDateTime": "2023-11-14T11:03:08.176Z",
    "endDateTime": "2024-08-14T11:03:08.176Z"
  },
  "actualCostItem": [
    {
      "@type": "ActualCostItem",
      "id": "123456",
      "description": "The actual cost of the fibre optic cable",
      "name": "Fiber Optic Cable",
      "creationDate": "2024-06-14T11:03:08.178Z",
      "lastUpdate": "2024-07-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
        }
      },
      "state": "incurred",
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "111000",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    }
  ]
}
```


Updates partially a ActualCost

`PATCH /actualCost/{id}?fields=…`

Description

This operation allows partial updates of an ActualCost entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

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Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
lastUpdate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
actualCostItem	
attachment	
costRelationship	
description	
externalIdentifier	
name	
note	
period	
relatedEntity	
relatedParty	
state	
version	

Usage samples

Here's an example of a request for patching an actual cost. In this example, a new description is set.

Request

```
PATCH /actualCost/19d72f32-13b8-4711-a805-12850ea1ce19
Content-Type: application/json
{
    "description" : "Costs incurred for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
    "@type": "ActualCost",
    "@baseType": "Cost"
}
```

Response

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```
200
{
  "@type": "ActualCost",
  "@baseType": "Cost",
  "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
  "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
  "name": "Actual Cost",
  "description": "Costs incurred for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
  "creationDate": "2023-11-14T11:03:08.176Z",
  "lastUpdate": "2024-08-14T11:03:08.176Z",
  "state": "inProgress",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Supplier",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "role": "initiator",
      "entity":
        {
          "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "@type": "EntityRef",
          "@referredType": "ServiceOrder"
        }
    }
  ],
  "period": {
    "startDateTime": "2023-11-14T11:03:08.176Z",
    "endDateTime": "2024-08-14T11:03:08.176Z"
  },
  "actualCostItem": [
    {
      "@type": "ActualCostItem",
      "id": "123456",
      "description": "The actual cost of the fibre optic cable",
      "name": "Fiber Optic Cable",
      "creationDate": "2024-06-14T11:03:08.178Z",
      "lastUpdate": "2024-07-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
```


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```
        }
      },
      "state": "allocated",
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "111000",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    },
    {
      "@type": "ActualCostItem",
      "id": "234567",
      "description": "The actual cost of the Electricity and Power Consumption",
      "name": "Electricity and Power Consumption",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 20000,
          "units": "kWh"
        },
        "amount": {
          "unit": "AUD",
          "value": 4000
        }
      },
      "state": "incurred",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      },
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "456456",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
```


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```
        }
      ]
    }
  ]
}
```

Here's an example of a request for patching an actual cost. In this example, a new description is set.

Request

```
PATCH /actualCost/19d72f32-13b8-4711-a805-12850ea1ce19
Content-Type: application/merge-patch+json
{
    "description" : "Costs incurred for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
    "@type": "ActualCost",
    "@baseType": "Cost"
}
```

Response

```
200
{
  "@type": "ActualCost",
  "@baseType": "Cost",
  "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
  "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
  "name": "Actual Cost",
  "description": "Costs incurred for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
  "creationDate": "2023-11-14T11:03:08.176Z",
  "lastUpdate": "2024-08-14T11:03:08.176Z",
  "state": "inProgress",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Supplier",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "role": "initiator",
      "entity":
        {
          "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "href": "https://mycsp.com:8080/tmf-
```

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```
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
          "@type": "EntityRef",
          "@referredType": "ServiceOrder"
        }
    }
  ],
  "period": {
    "startDateTime": "2023-11-14T11:03:08.176Z",
    "endDateTime": "2024-08-14T11:03:08.176Z"
  },
  "actualCostItem": [
    {
      "@type": "ActualCostItem",
      "id": "123456",
      "description": "The actual cost of the fibre optic cable",
      "name": "Fiber Optic Cable",
      "creationDate": "2024-06-14T11:03:08.178Z",
      "lastUpdate": "2024-07-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
        }
      },
      "state": "allocated",
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "111000",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    },
    {
      "@type": "ActualCostItem",
      "id": "234567",
      "description": "The actual cost of the Electricity and Power Consumption",
      "name": "Electricity and Power Consumption",
      "creationDate": "2024-07-14T11:03:08.178Z",
      "lastUpdate": "2024-08-14T11:03:08.178Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 20000,
          "units": "kWh"
        },
```


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```
        "amount": {
          "unit": "AUD",
          "value": 4000
        }
      },
      "state": "incurred",
      "period": {
        "startDateTime": "2024-07-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      },
      "costItemRelationship": [
        {
          "@type": "CostItemRelationship",
          "@baseType": "CostItemRelationship",
          "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
          "costItemRelationshipCharacteristic": [],
          "costItem": {
            "@type": "CostItemRef",
            "@baseType": "CostItemRef",
            "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
            "@referredType": "projectedCostItem",
            "itemId": "456456",
            "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
            "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
          },
          "relationshipType": "projectedBy"
        }
      ]
    }
  ]
}
```


Here's an example of a request for patching an actual cost. In this example, an external identifier is added.

Request

```
PATCH /actualCost/19d72f32-13b8-4711-a805-12850ea1ce19
Content-Type: application/json-patch+json
[{
    "op": "add",
    "path": "/externalIdentifier",
    "value": {
        "id": "FIN-12345",
        "externalIdentifierType":"Auditing Report",
        "@type": "ExternalIdentifier"
    }
}]
```

Response


```
200
{
    "@type": "ActualCost",
    "@baseType": "Cost",
    "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
```


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```
13b8-4711-a805-12850ea1ce19",
    "id": "19d72f32-13b8-4711-a805-12850ea1ce19",
    "name": "Actual Cost",
    "creationDate": "2023-11-14T11:03:08.176Z",
    "lastUpdate": "2024-08-14T11:03:08.176Z",
    "state": "inProgress",
    "externalIdentifier":[
        {
            "id": "FIN-12345",
            "externalIdentifierType":"Auditing Report",
            "@type": "ExternalIdentifier"
        }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
          }
      }
    ],
    "period": {
      "startDateTime": "2023-11-14T11:03:08.176Z",
      "endDateTime": "2024-08-14T11:03:08.176Z"
    },
    "actualCostItem": [
      {
        "@type": "ActualCostItem",
        "id": "123456",
        "description": "The actual cost of the fibre optic cable",
        "name": "Fiber Optic Cable",
        "creationDate": "2024-06-14T11:03:08.178Z",
        "lastUpdate": "2024-07-14T11:03:08.178Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10,
            "units": "km"
          },
          "amount": {
            "unit": "AUD",
            "value": 50000
          }
```


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```
        },
        "state": "allocated",
        "costItemRelationship": [
          {
            "@type": "CostItemRelationship",
            "@baseType": "CostItemRelationship",
            "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
            "costItemRelationshipCharacteristic": [],
            "costItem": {
              "@type": "CostItemRef",
              "@baseType": "CostItemRef",
              "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
              "@referredType": "projectedCostItem",
              "itemId": "111000",
              "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
              "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
            },
            "relationshipType": "projectedBy"
          }
        ]
      },
      {
        "@type": "ActualCostItem",
        "id": "234567",
        "description": "The actual cost of the Electricity and Power Consumption",
        "name": "Electricity and Power Consumption",
        "creationDate": "2024-07-14T11:03:08.178Z",
        "lastUpdate": "2024-08-14T11:03:08.178Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 20000,
            "units": "kWh"
          },
          "amount": {
            "unit": "AUD",
            "value": 4000
          }
        },
        "state": "incurred",
        "period": {
          "startDateTime": "2024-07-14T11:03:08.176Z",
          "endDateTime": "2024-08-14T11:03:08.176Z"
        },
        "costItemRelationship": [
          {
            "@type": "CostItemRelationship",
            "@baseType": "CostItemRelationship",
            "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
            "costItemRelationshipCharacteristic": [],
            "costItem": {
              "@type": "CostItemRef",
              "@baseType": "CostItemRef",
              "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
              "@referredType": "projectedCostItem",
              "itemId": "456456",
              "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
              "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
            },
            "relationshipType": "projectedBy"
          }
```


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```
        ]
      }
    ]
  }
```

Here's an example of a request for patching an actual cost. In this example, the state is updated for acutalCostItem.id=123456

Request

```
PATCH /actualCost/19d72f32-13b8-4711-a805-12850ea1ce19
Content-Type: application/json-patch-query+json

  [{
      "op": "replace",
      "path": "/actualCostItem/state?actualCostItem.id=123456",
      "value":"verified"
    }]
```

Response

```
200
{
    "@type": "ActualCost",
    "@baseType": "Cost",
    "id":"19d72f32-13b8-4711-a805-12850ea1ce19",
    "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/actualCost/19d72f32-
13b8-4711-a805-12850ea1ce19",
    "name": "Actual Cost",
    "creationDate": "2023-11-14T11:03:08.176Z",
    "lastUpdate": "2024-08-14T11:03:08.176Z",
    "state": "inProgress",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
```

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```
          }
        }
      ],
      "period": {
        "startDateTime": "2023-11-14T11:03:08.176Z",
        "endDateTime": "2024-08-14T11:03:08.176Z"
      },
      "actualCostItem": [
        {
          "@type": "ActualCostItem",
          "id": "123456",
          "description": "The actual cost of the fibre optic cable",
          "name": "Fiber Optic Cable",
          "creationDate": "2024-06-14T11:03:08.178Z",
          "lastUpdate": "2024-07-14T11:03:08.178Z",
          "isTaxIncluded": true,
          "costItemAmount": {
            "quantity": {
              "amount": 10,
              "units": "km"
            },
            "amount": {
              "unit": "AUD",
              "value": 50000
            }
          },
          "state": "verified",
          "costItemRelationship": [
            {
              "@type": "CostItemRelationship",
              "@baseType": "CostItemRelationship",
              "@schemaLocation": "http://my.schemas/CostItemRelationship.schema.json",
              "costItemRelationshipCharacteristic": [],
              "costItem": {
                "@type": "CostItemRef",
                "@baseType": "CostItemRef",
                "@schemaLocation": "http://my.schemas/CostItemRef.schema.json",
                "@referredType": "projectedCostItem",
                "itemId": "111000",
                "costHref": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
                "costId": "19d72f32-13b8-4711-a805-12850ea1ce20"
              },
              "relationshipType": "projectedBy"
            }
          ]
        }
      ]
    }
```


Deletes a ActualCost

`DELETE /actualCost/{id}`

Description

This operation deletes an ActualCost entity.

Usage samples

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Here's an example of a request for deleting an actual cost.

Request

```
DELETE /actualCost/19d72f32-13b8-4711-a805-12850ea1ce19
Content-Type: application/json
```

Response

```
204
```

Operations on ProjectedCost

Retrieves a ProjectedCost by ID

`GET /projectedCost/{id}?fields=…`

Description

This operation retrieves a ProjectedCost entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific projected cost instance.

Request

```
GET /projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20
Content-Type: application/json
```

Response

```
200
{
  "@type": "ProjectedCost",
  "@baseType": "Cost",
  "href": "https://mycsp.com:8080/tmf-api/costManagement/v5/projectedCost/19d72f32-
13b8-4711-a805-12850ea1ce20",
  "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
  "description": "Cost projection for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
  "name": "Projected Cost",
  "creationDate": "2024-08-15T14:20:34.396Z",
  "lastUpdate": "2024-08-15T14:20:34.396Z",
  "state": "pendingApproval",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Supplier",
```

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```
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
        "id": "456",
        "name": "NetCo Pty Ltd",
        "@referredType": "Organization"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "role": "initiator",
      "entity": {
        "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
        "@type": "EntityRef",
        "@referredType": "ServiceOrder"
      }
    }
  ],
  "certaintyLevel": "Firm cost",
  "validFor": {
    "startDateTime": "2024-08-15T14:20:34.396Z",
    "endDateTime": "2024-09-15T14:20:34.396Z"
  },
  "projectedCostItem": [
    {
      "@type": "ProjectedCostItem",
      "id": "898989",
      "description": "E-line connectivity service",
      "name": "E-line",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
          "value": 500
        }
      },
      "tax": [{
        "taxCategory": "GST",
        "taxRate": 0.10,
        "@type":"TaxItem"
      }],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "costItemType": "recurring"
    },
    {
      "@type": "ProjectedCostItem",
      "id": "111000",
      "description": "The projected cost of the fibre optic cable",
```


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```
      "name": "Fiber Optic Cable",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 10,
          "units": "km"
        },
        "amount": {
          "unit": "AUD",
          "value": 50000
        }
      },
      "tax": [{
        "taxCategory": "GST",
        "taxRate": 0.10,
        "@type":"TaxItem"
      }],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T23:20:50.52Z",
        "endDateTime": "2024-09-15T23:20:50.52Z"
      },
      "costItemType": "oneTime"
    },
    {
      "@type": "ProjectedCostItem",
      "id": "456456",
      "description": "The projected cost of the Electricity and Power Consumption",
      "name": "Electricity and Power Consumption",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 10000,
          "units": "kWh"
        },
        "amount": {
          "unit": "AUD",
          "value": 2000
        }
      },
      "tax":[{
        "taxCategory": "GST",
        "taxRate": 0.10,
        "@type":"TaxItem"
      }],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "costItemType": "consumptionBased"
    },
    {
      "@type": "ProjectedCostItem",
      "id": "456458",
      "description": "Maintenance and support",
      "name": "Maintenance and support",
      "creationDate": "2024-08-15T14:20:34.396Z",
      "lastUpdate": "2024-08-15T14:20:34.396Z",
```


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```
      "isTaxIncluded": true,
      "costItemAmount": {
        "quantity": {
          "amount": 1,
          "units": "month"
        },
        "amount": {
          "unit": "AUD",
          "value": 200
        }
      },
      "tax": [{
        "taxCategory": "GST",
        "taxRate": 0.10,
        "@type":"TaxItem"
      }],
      "state": "estimated",
      "validFor": {
        "startDateTime": "2024-08-15T14:20:34.396Z",
        "endDateTime": "2024-09-15T14:20:34.396Z"
      },
      "costItemType": "recurring"
    }
  ]
}
```


List or find ProjectedCost objects

`GET /projectedCost?fields=…`

Description

This operation list ProjectedCost entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple projected cost instances, in this example approved projected actual cost instances are filtered and id, href, name and state fields need to be returned

Request

```
GET /projectedCost?state=approved&fields=id,name,state,href
Content-Type: application/json
```

Response


```
200
[
    {
      "name": "Projected Cost",
      "id":"096d29a9-7bcc-4fca-8402-1c10c4d15cbb",
      "href": "https://mycsp.com:8080/tmf-
```


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```
api/costManagement/v5/projectedCost/096d29a9-7bcc-4fca-8402-1c10c4d15cbb",
      "state":"approved",
      "@type": "ProjectedCost"
    },
    {
      "name": "Projected Cost",
      "id":"3e0d5949-c928-4b64-b4e8-1723f3364fa4",
      "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/3e0d5949-c928-4b64-b4e8-1723f3364fa4",
      "state":"approved",
      "@type": "ProjectedCost"
    }
  ]
```

Creates a ProjectedCost

POST /projectedCost?fields=…

Description

This operation creates a ProjectedCost entity.

Mandatory Attributes

Mandatory Attributes	Rule
name	
projectedCostItem	
@type	

Usage samples

Here's an example of a request for creating an projected cost

Request

```
POST /projectedCost
Content-Type: application/json
{
    "@type": "ProjectedCost",
    "@baseType": "Cost",
    "description": "Fibre cost projection for Service Order SO1",
    "name": "Projected Cost",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role":  "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
```

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```
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
          }
      }
    ],
    "certaintyLevel": "Firm cost",
    "validFor": {
      "startDateTime": "2024-08-15T14:20:34.396Z",
      "endDateTime": "2024-09-15T14:20:34.396Z"
    },
    "projectedCostItem": [
      {
        "@type": "ProjectedCostItem",
        "id": "111000",
        "description": "The projected cost of the fibre optic cable",
        "name": "Fiber Optic Cable",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10,
            "units": "km"
          },
          "amount": {
            "unit": "AUD",
            "value": 50000
          }
        },
        "state": "estimated",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "oneTime"
      },
      {
        "@type": "ProjectedCostItem",
        "id": "456456",
        "description": "The projected cost of the Electricity and Power
Consumption",
        "name": "Electricity and Power Consumption",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10000,
            "units": "kWh"
          },
          "amount": {
            "unit": "AUD",
            "value": 2000
          }
        },
        "state": "estimated",
        "validFor": {
```


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```
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "consumptionBased"
      }
    ]
  }
```

Response

```
201
{
    "@type": "ProjectedCost",
    "@baseType": "Cost",
    "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
    "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
    "description": "Fibre cost projection for Service Order SO1",
    "name": "Projected Cost",
    "creationDate": "2024-08-15T14:20:34.396Z",
    "lastUpdate": "2024-08-15T14:20:34.396Z",
    "state": "inProgress",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role":  "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
          }
      }
    ],
    "certaintyLevel": "Firm cost",
    "validFor": {
      "startDateTime": "2024-08-15T14:20:34.396Z",
      "endDateTime": "2024-09-15T14:20:34.396Z"
    },
    "projectedCostItem": [
      {
        "@type": "ProjectedCostItem",
        "id": "111000",
        "description": "The projected cost of the fibre optic cable",
```

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```
        "name": "Fiber Optic Cable",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10,
            "units": "km"
          },
          "amount": {
            "unit": "AUD",
            "value": 50000
          }
        },
        "state": "estimated",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "oneTime"
      },
      {
        "@type": "ProjectedCostItem",
        "id": "456456",
        "description": "The projected cost of the Electricity and Power
Consumption",
        "name": "Electricity and Power Consumption",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10000,
            "units": "kWh"
          },
          "amount": {
            "unit": "AUD",
            "value": 2000
          }
        },
        "state": "estimated",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "consumptionBased"
      }
    ]
  }
```


Updates partially a ProjectedCost

`PATCH /projectedCost/{id}?fields=…`

Description

This operation allows partial updates of a ProjectedCost entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default

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value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
lastUpdate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
certaintyLevel	
costRelationship	
description	
externalIdentifier	
name	
note	
projectedCostItem	
relatedEntity	
relatedParty	
state	
validFor	
version	

Usage samples

Here's an example of a request for patching an projected cost. In this example, a new description is set.

Request

```
PATCH /projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20
Content-Type: application/json
{
    "description" : "Projected costs for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
    "@type": "ProjectedCost",
```

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```
    "@baseType": "Cost"
}
```

Response

```
200
{
    "@type": "ProjectedCost",
    "@baseType": "Cost",
    "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
    "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
    "description": "Projected costs for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
    "name": "Projected Cost",
    "creationDate": "2024-08-15T14:20:34.396Z",
    "lastUpdate": "2024-08-15T14:20:34.396Z",
    "state": "inProgress",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role":  "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
          }
      }
    ],
    "certaintyLevel": "Firm cost",
    "validFor": {
      "startDateTime": "2024-08-15T14:20:34.396Z",
      "endDateTime": "2024-09-15T14:20:34.396Z"
    },
    "projectedCostItem": [
      {
        "@type": "ProjectedCostItem",
        "id": "111000",
        "description": "The projected cost of the fibre optic cable",
        "name": "Fiber Optic Cable",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
```

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```
        "costItemAmount": {
          "quantity": {
            "amount": 10,
            "units": "km"
          },
          "amount": {
            "unit": "AUD",
            "value": 50000
          }
        },
        "state": "approved",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "oneTime"
      },
      {
        "@type": "ProjectedCostItem",
        "id": "456456",
        "description": "The projected cost of the Electricity and Power
Consumption",
        "name": "Electricity and Power Consumption",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10000,
            "units": "kWh"
          },
          "amount": {
            "unit": "AUD",
            "value": 2000
          }
        },
        "state": "approved",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "consumptionBased"
      }
    ]
  }
```


Here's an example of a request for patching an projected cost. In this example, a new description is set.

Request

```
PATCH /projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20
Content-Type: application/merge-patch+json
{
    "description" : "Projected costs for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
    "@type": "ProjectedCost",
    "@baseType": "Cost"
}
```

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Response


```
200
{
    "@type": "ProjectedCost",
    "@baseType": "Cost",
    "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
    "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
    "description": "Projected costs for Service Order 3f845e07-73d8-469b-949f-
fe95e52ab5ab",
    "name": "Projected Cost",
    "creationDate": "2024-08-15T14:20:34.396Z",
    "lastUpdate": "2024-08-15T14:20:34.396Z",
    "state": "inProgress",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role":  "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
          }
      }
    ],
    "certaintyLevel": "Firm cost",
    "validFor": {
      "startDateTime": "2024-08-15T14:20:34.396Z",
      "endDateTime": "2024-09-15T14:20:34.396Z"
    },
    "projectedCostItem": [
      {
        "@type": "ProjectedCostItem",
        "id": "111000",
        "description": "The projected cost of the fibre optic cable",
        "name": "Fiber Optic Cable",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10,
            "units": "km"
```


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```
          },
          "amount": {
            "unit": "AUD",
            "value": 50000
          }
        },
        "state": "approved",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "oneTime"
      },
      {
        "@type": "ProjectedCostItem",
        "id": "456456",
        "description": "The projected cost of the Electricity and Power
Consumption",
        "name": "Electricity and Power Consumption",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10000,
            "units": "kWh"
          },
          "amount": {
            "unit": "AUD",
            "value": 2000
          }
        },
        "state": "approved",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "consumptionBased"
      }
    ]
  }
```


Here's an example of a request for patching an projected cost. In this example, an external identifier is added.

Request

```
PATCH /projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20
Content-Type: application/json-patch+json
[{
    "op": "add",
    "path": "/externalIdentifier",
    "value": {
        "id": "FIN-12345",
        "externalIdentifierType":"Auditing Report",
        "@type": "ExternalIdentifier"
    }
}]
```

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Response


```
200
{
    "@type": "ProjectedCost",
    "@baseType": "Cost",
    "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
    "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
    "description": "Fibre cost projection for Service Order SO1",
    "name": "Projected Cost",
    "creationDate": "2024-08-15T14:20:34.396Z",
    "lastUpdate": "2024-08-15T14:20:34.396Z",
    "state": "inProgress",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role":  "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
          }
      }
    ],
    "externalIdentifier":[
        {
            "id": "FIN-12345",
            "externalIdentifierType":"Auditing Report",
            "@type": "ExternalIdentifier"
        }
    ],
    "certaintyLevel": "Firm cost",
    "validFor": {
      "startDateTime": "2024-08-15T14:20:34.396Z",
      "endDateTime": "2024-09-15T14:20:34.396Z"
    },
    "projectedCostItem": [
      {
        "@type": "ProjectedCostItem",
        "id": "111000",
        "description": "The projected cost of the fibre optic cable",
        "name": "Fiber Optic Cable",
        "creationDate": "2024-08-15T14:20:34.396Z",
```


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```
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10,
            "units": "km"
          },
          "amount": {
            "unit": "AUD",
            "value": 50000
          }
        },
        "state": "approved",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "oneTime"
      },
      {
        "@type": "ProjectedCostItem",
        "id": "456456",
        "description": "The projected cost of the Electricity and Power
Consumption",
        "name": "Electricity and Power Consumption",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10000,
            "units": "kWh"
          },
          "amount": {
            "unit": "AUD",
            "value": 2000
          }
        },
        "state": "approved",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "consumptionBased"
      }
    ]
  }
```


Here's an example of a request for patching an projected cost. In this example, the state is updated for acutalCostItem.id=123456

Request

```
PATCH /projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20
Content-Type: application/json-patch-query+json
[{
    "op": "replace",
    "path": "/actualCostItem/state?actualCostItem.id=111000",
    "value":"approved"
```

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```
  }]
```

Response

```
200
{
    "@type": "ProjectedCost",
    "@baseType": "Cost",
    "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20",
    "id": "19d72f32-13b8-4711-a805-12850ea1ce20",
    "description": "Fibre cost projection for Service Order SO1",
    "name": "Projected Cost",
    "creationDate": "2024-08-15T14:20:34.396Z",
    "lastUpdate": "2024-08-15T14:20:34.396Z",
    "state": "inProgress",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role":  "Supplier",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/456",
          "id": "456",
          "name": "NetCo Pty Ltd",
          "@referredType": "Organization"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "role": "initiator",
        "entity":
          {
            "id": "3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "href": "https://mycsp.com:8080/tmf-
api/serviceOrdering/v5/serviceOrder/3f845e07-73d8-469b-949f-fe95e52ab5ab",
            "@type": "EntityRef",
            "@referredType": "ServiceOrder"
          }
      }
    ],
    "certaintyLevel": "Firm cost",
    "validFor": {
      "startDateTime": "2024-08-15T14:20:34.396Z",
      "endDateTime": "2024-09-15T14:20:34.396Z"
    },
    "projectedCostItem": [
      {
        "@type": "ProjectedCostItem",
        "id": "111000",
        "description": "The projected cost of the fibre optic cable",
        "name": "Fiber Optic Cable",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
```

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```
            "amount": 10,
            "units": "km"
          },
          "amount": {
            "unit": "AUD",
            "value": 50000
          }
        },
        "state": "approved",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "oneTime"
      },
      {
        "@type": "ProjectedCostItem",
        "id": "456456",
        "description": "The projected cost of the Electricity and Power
Consumption",
        "name": "Electricity and Power Consumption",
        "creationDate": "2024-08-15T14:20:34.396Z",
        "lastUpdate": "2024-08-15T14:20:34.396Z",
        "isTaxIncluded": true,
        "costItemAmount": {
          "quantity": {
            "amount": 10000,
            "units": "kWh"
          },
          "amount": {
            "unit": "AUD",
            "value": 2000
          }
        },
        "state": "estimated",
        "validFor": {
          "startDateTime": "2024-08-15T14:20:34.396Z",
          "endDateTime": "2024-09-15T14:20:34.396Z"
        },
        "costItemType": "consumptionBased"
      }
    ]
  }
```


Deletes a ProjectedCost

`DELETE /projectedCost/{id}`

Description

This operation deletes a ProjectedCost entity.

Usage samples

Here's an example of a request for deleting a projected cost.

Request

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```
DELETE /projectedCost/19d72f32-13b8-4711-a805-12850ea1ce20
Content-Type: application/json
```

Response

```
204
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0	Jan 2025	Anu Aulakh Telstra anu.aulakh@team.telstra.com	First release
Version 5.0.0	Jan 2025	Amita Giriya Telstra amita.giriya@team.telstra.com	First release
Version 5.0.0	Jan 2025	Varun Sasidharan Nair Telstra varun.nair@team.telstra.com	First release
Version 5.0.0	Jan 2025	Abdul Majid Hussain Telstra abdul.majidhussain@team.telstra.com	First release

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